

Supplementary Material

Multi-objective Pareto Hyperparameter Optimization for Joint Detection and Localization of Forged Identity Documents

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This document is a companion to the Pattern Recognition Letters submission *Multi-objective Pareto Hyperparameter Optimization for Joint Detection and Localization of Forged Identity Documents* (DocVerify). It collects the complete experimental tables that, owing to the journal’s strict page limit, are summarized in only one or two sentences in the main paper or in the response to reviewers. All numbers are computed from the released result CSVs (archived together with this document); section references (e.g. Sec. 4.3) point to the main paper. Unless stated otherwise, all blind-test results use the held-out 15% FantasyID partition, 224×224 inputs, and the Pareto-selected configuration of Table 2 in the main paper. Statistical tests are paired two-sided Wilcoxon signed-rank tests with Holm correction across metrics within an experiment; effect sizes are paired Cohen’s d .

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S1 Computational cost breakdown

The ~ 59 -hour budget reported in Sec. 4.1 is the cost of the full *scientific* protocol (unbiased nested cross-validation plus a 30-seed ablation), not the cost of *using* the method. Table [S1](#) gives the measured wall-clock breakdown on a single NVIDIA RTX 5060 Ti (16 GB) with VRAM caching. Once the configuration is fixed, retraining the single deployed model from scratch (up to 100 epochs with early stopping) costs only 7.7 ± 1.0 min per seed; the Pareto loss weighting adds no inference-time cost.

S2 Data-augmentation ablation

Sec. 4.1 states that standard augmentation degraded every metric on the FantasyID holdout. Table [S2](#) gives the full 30-seed comparison. The augmented model applies, in the training loader

Table S1: Wall-clock cost per phase (single GPU, VRAM-cached). Per-seed retraining time is the mean over the 30 blind-test seeds (range 6.0–10.2 min).

Phase	Trainings	Wall-clock
Nested HPO search (10 outer $\times$ 50 MOTPE $\times$ 5 inner)	2 500	~ 50 h
Outer final models (one retrain per outer fold)	10	~ 1 h
Blind-test ablation (30 seeds $\times$ 4 variants)	120	~ 12 h
Total scientific protocol	2 630	~ 59–63 h
Practical deployment (one fixed config, one seed)	1	7.7 ± 1.0 min

only, random rotation ($\pm 10^\circ$), brightness, contrast, Gaussian blur, and JPEG compression, using the *same* Pareto configuration. On the clean synthetic holdout, augmentation significantly degrades every metric (all Holm $p < 10^{-7}$) and inflates variance by 2–5 $\times$, confirming the paper’s no-augmentation choice for this dataset.

Table S2: Augmentation ablation (blind test, 30 seeds). Δ : aug – no_aug; d : paired Cohen’s d ; p_{Holm} : Holm-corrected paired Wilcoxon. All differences significant.

Metric	No aug (ours)	Aug	Δ	d	p_{Holm}
PR-AUC	0.9967 ± 0.0021	0.9577 ± 0.0457	-0.0390	-0.84	1.5×10^{-8}
Dice	0.8750 ± 0.0180	0.7797 ± 0.0396	-0.0953	-2.10	9.3×10^{-9}
BAcc	0.9573 ± 0.0184	0.8302 ± 0.1007	-0.1271	-1.20	5.6×10^{-9}
F1 ₁	0.9651 ± 0.0152	0.8593 ± 0.1061	-0.1058	-0.97	1.1×10^{-8}
F1 _{macro}	0.9423 ± 0.0234	0.7998 ± 0.1136	-0.1425	-1.19	7.5×10^{-9}

S3 Class-imbalance (pos_weight) sensitivity

Sec. 3.1 notes that the mild class imbalance (2.5:1, attack-majority) leaves PR-AUC robust and that a `pos_weight` sweep confirms no significant effect. Table S3 reports the three settings (inverse-frequency 0.397, neutral 1.0, and the attack-favouring 2.518), each over 30 seeds. No metric differs significantly from the neutral baseline after Holm correction (all $p_{\text{Holm}} > 0.5$), so no loss re-weighting is applied in the main model.

Table S3: `pos_weight` sensitivity (blind test, 30 seeds). Baseline is `pos_weight`= 1.0. No comparison against the baseline reaches significance after Holm correction.

<code>pos_weight</code>	PR-AUC	Dice	BAcc	Rec. attack	Rec. bonafide
0.397 (inv. freq.)	0.9973 ± 0.0018	0.8793 ± 0.0168	0.9614 ± 0.0146	0.9431	0.9797
1.0 (baseline)	0.9967 ± 0.0021	0.8750 ± 0.0180	0.9573 ± 0.0184	0.9435	0.9712
2.518 (attack-favouring)	0.9967 ± 0.0019	0.8688 ± 0.0240	0.9552 ± 0.0164	0.9435	0.9669

S4 Selection criterion: Pareto vs. scalar (full results)

Table 3 of the main paper reports a compact form of the selection-criterion study. Table S4 gives the complete 30-seed results with 95% confidence intervals for all five metrics, and Table S5 the

paired statistics. Both scalar criteria select the global argmax (by validation PR-AUC or validation Dice) over the *same* 500 MOTPE trials; our Pareto criterion uses the distance to the ideal point. Pareto improves localization over **by_prauc** and improves *both* objectives over **by_dice**.

Table S4: Selection-criterion comparison (blind test, 30 seeds). Mean $\pm$ Std with 95% CI in brackets.

Metric	Pareto (ours)	by_prauc	by_dice
PR-AUC	0.9967 ± 0.0021 [0.9959, 0.9975]	0.9953 ± 0.0021 [0.9945, 0.9961]	0.9948 ± 0.0027 [0.9938, 0.9958]
Dice	0.8750 ± 0.0180 [0.8683, 0.8818]	0.8589 ± 0.0230 [0.8504, 0.8675]	0.8566 ± 0.0147 [0.8511, 0.8620]
BAcc	0.9573 ± 0.0184 [0.9504, 0.9642]	0.9481 ± 0.0146 [0.9427, 0.9536]	0.9463 ± 0.0170 [0.9400, 0.9527]
F1 ₁	0.9651 ± 0.0152 [0.9594, 0.9708]	0.9569 ± 0.0122 [0.9523, 0.9614]	0.9564 ± 0.0121 [0.9519, 0.9609]
F1 _{macro}	0.9423 ± 0.0234 [0.9335, 0.9510]	0.9294 ± 0.0184 [0.9225, 0.9362]	0.9283 ± 0.0184 [0.9214, 0.9352]

Table S5: Paired statistics, Pareto vs. each scalar criterion (30 seeds). Δ : Pareto – scalar; d : paired Cohen’s d ; p_{Holm} : Holm-corrected paired Wilcoxon; $\checkmark$: significant at $p_{\text{Holm}} < 0.05$.

Comparison	Metric	Δ	d	p_{Holm}	sig.
Pareto vs by_prauc	PR-AUC	+0.0014	0.49	0.062	
	Dice	+0.0161	0.59	0.043	$\checkmark$
	BAcc	+0.0092	0.40	0.061	
	F1 ₁	+0.0082	0.45	0.066	
	F1 _{macro}	+0.0129	0.46	0.050	$\checkmark$
Pareto vs by_dice	PR-AUC	+0.0019	0.64	0.0072	$\checkmark$
	Dice	+0.0185	1.00	0.00039	$\checkmark$
	BAcc	+0.0110	0.45	0.040	$\checkmark$
	F1 ₁	+0.0087	0.44	0.066	
	F1 _{macro}	+0.0140	0.47	0.043	$\checkmark$

S5 Architecture-agnostic validation (ResNet-18)

Sec. 4.6 states that the identical MOTPE protocol, applied to an ImageNet-pretrained ResNet-18 backbone, yields comparable blind-test performance and that neither model dominates. Table S6 gives the full 30-seed comparison. The ResNet-18 Pareto-selected configuration (distance to ideal = 0.164) is $\eta = 3.68 \times 10^{-4}$, $\lambda_{\text{wd}} = 1.45 \times 10^{-6}$, dropout = 0.381, $C_{\text{dec}} = 256$, $\lambda_{\text{mask}} = 2.69$, the last consistent with DocVerify’s 2.46. DocVerify is better on PR-AUC; ResNet-18 is better on Dice and the F1 metrics; balanced accuracy does not differ significantly. The contribution is therefore the multi-objective selection methodology, not a specific encoder.

S6 Cross-domain generalization (SIDTD)

Sec. 4.6 reports that the FantasyID-trained model is near chance zero-shot on the out-of-domain SIDTD dataset, but recovers strong performance when the same architecture is retrained on the SIDTD training partition. Table S7 gives the full curve from zero-shot ($n_{\text{shots}} = 0$) through few-shot to full retraining ($n_{\text{shots}} = 1777$). Few-shot adaptation up to 400 samples stays near chance;

Table S6: DocVerify vs. ResNet-18 under the identical MOTPE protocol (blind test, 30 seeds). Δ : DocVerify – ResNet-18; d : paired Cohen’s d ; p_{Holm} : Holm-corrected paired Wilcoxon.

Metric	DocVerify	ResNet18-MOTPE	Δ	d	p_{Holm}
PR-AUC	0.9967 ± 0.0021	0.9942 ± 0.0025	+0.0024	+0.75	2.8×10^{-4}
Dice	0.8750 ± 0.0180	0.8894 ± 0.0111	−0.0144	−0.60	8.1×10^{-3}
BAcc	0.9573 ± 0.0184	0.9629 ± 0.0108	−0.0055	−0.21	0.285 (n.s.)
F1 _l	0.9651 ± 0.0152	0.9779 ± 0.0069	−0.0128	−0.67	3.8×10^{-3}
F1 _{macro}	0.9423 ± 0.0234	0.9613 ± 0.0114	−0.0190	−0.63	4.0×10^{-3}

only full in-domain retraining recovers performance, confirming that the gap reflects domain shift rather than architectural capacity.

Table S7: SIDTD adaptation curve. $n_{\text{shots}} = 0$ is zero-shot (30 seeds); $n_{\text{shots}} \geq 10$ are few-shot/full retrains (25 seeds each). Templates subset (1 000 bonafide, 1 222 attack).

n_{shots}	PR-AUC	ROC-AUC	BAcc	F1 _{macro}
0 (zero-shot)	0.5554 ± 0.0110	0.5041 ± 0.0094	0.5010	—
10	0.5485 ± 0.0084	0.4893 ± 0.0147	0.5002 ± 0.0135	0.4841 ± 0.0199
25	0.5595 ± 0.0089	0.4996 ± 0.0092	0.5046 ± 0.0146	0.4755 ± 0.0255
50	0.5575 ± 0.0102	0.4960 ± 0.0128	0.4953 ± 0.0167	0.4652 ± 0.0207
100	0.5706 ± 0.0208	0.5060 ± 0.0199	0.5032 ± 0.0203	0.4848 ± 0.0259
200	0.5774 ± 0.0170	0.5089 ± 0.0144	0.5031 ± 0.0182	0.4853 ± 0.0289
400	0.5912 ± 0.0238	0.5123 ± 0.0171	0.5028 ± 0.0184	0.4843 ± 0.0349
1 777 (full)	0.8780 ± 0.0161	0.8263 ± 0.0228	0.7188 ± 0.0349	0.7142 ± 0.0333

S7 Augmentation does not close the cross-domain gap

As a follow-up to Sections S2 and S6, we tested whether augmentation-trained checkpoints generalize better zero-shot to SIDTD. Table S8 shows a null result: both the augmented and non-augmented models remain at chance, with no significant difference in detection quality after Holm correction. Augmentation therefore does not substitute for in-domain data; closing the gap requires multi-source training (future work, Sec. 6).

Table S8: Zero-shot SIDTD generalization, augmented vs. non-augmented FantasyID training (30 seeds each). Δ : aug – no_aug; p_{Holm} : Holm-corrected paired Wilcoxon.

Metric	No aug	Aug	Δ	p_{Holm}
PR-AUC	0.5554 ± 0.0110	0.5639 ± 0.0096	+0.0085	0.077 (n.s.)
ROC-AUC	0.5041 ± 0.0094	0.5089 ± 0.0060	+0.0048	0.199 (n.s.)

Data availability. The result CSVs underlying every table above, together with the experiment scripts, are released in the reproducibility archive accompanying this document.